Toward a four-dimensional Yang–Mills construction

Notebook 36: soft determinants and bounded coarse corrections

Research working document

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Abstract

We factor the actual covariant mean-zero determinant of Notebook 35 into a soft scalar/vector pair and two determinants with one color fiber per final block. Both coarse determinants have uniformly bounded actual Maxwell coefficients as the block size grows. Their flat gaps alone would not prove this: we establish rooted physical source bounds, with all powers of the fine spacing and all fiber dimensions retained.

A new gauge-covariant Hessian estimate controls the actual effective classical action through its first two background derivatives, uniformly in preceding depth. Local formal path gauges isolate curvature; an ordered Laurent factorization retains the shifted lattice components. This yields the relative vertices needed for the vector correction. The auxiliary off-shell family is identified with the actual one only through the proved on-shell affine jets.

The fine soft pair has uniformly bounded short-time and long-time affine heat contributions. The latter follows from a local cubic resolvent formula and Schatten bounds that retain the block projector. Consequently only the intermediate heat interval can supply the required logarithm.

The resulting reduction retains the full effective action and original constraint normalization. It does not yet compute the remaining fine determinant's logarithmic coefficient or establish continuum Yang–Mills existence.

Conventions and proof boundary

The primitive odd ratio ℓ is fixed and $K = \ell^k$. Current coordinates use counting pairings; their spacing in final physical units is K^{-1}. Normalization $\mathcal{N}_0$ subtracts the flat value, and $[t^2]$ means a Taylor coefficient. All actual identities refer to the homogeneous stationary germ constructed in the preceding notebooks. Exact finite determinant identities and finite flat spectral estimates are distinguished from uniform spatial and curvature estimates on a chosen joint infinite classical structural system. No uniqueness of that system is assumed.

For a trace-normalized Cartan element H, put

$$C_H = \sum_{q \in \mathrm{spec}(\mathrm{ad}_H)} q^2, \qquad \alpha_g = \frac{C_H}{48\pi^2}.$$

The target remains the actual coefficient $22\alpha_g \log K + O(1)$, uniformly in preceding depth. The finite-part estimates below advance that comparison; they do not substitute a narrower objective for the existence problem.

1 A soft scalar factorization with a bounded coarse correction

Set $\mu = K^{-1}$. All fine scalar and bond pairings use counting measure. In this section the current links are $U(e) = \exp(i\mu A(e))$, so A is a physical potential in final units. Use the actual covariant

moving-reference block average B of Notebook 35, and write

$$J = K^2 B^T, \quad J^T J = I, \quad \Pi = J J^T, \quad P = I - \Pi, \quad \Delta = D^T D. \tag{1}$$

The normalized block column J_y has K^4 entries of size K^{-2}, each an adjoint transport. Distinct columns have disjoint site supports. Its background dependence stays inside its own block. Define

$$T = \Delta + \mu^2 \Pi, \qquad M = \mu^2 J^T T^{-1} J, \qquad \widehat{G} = P \Delta P + \mu^2 \Pi. \tag{2}$$

The letter M in this section denotes this coarse scalar matrix; preceding homogeneous depth will be written M_{old}.

Theorem 1.1 (Exact scalar factorization)**.** *On the actual local germ, continued through flatness,*

$$\det \widehat{G} = \det T \, \det M, \qquad \mathcal{N}_0 \log \det \widehat{G} = \mathcal{N}_0 \log \det T + \mathcal{N}_0 \log \det M. \tag{3}$$

Both factors are gauge covariant and have positive flat spectra. Their physical complex domain can be chosen independently of K. Moreover,

$$|\beta(\mathcal{N}_0 \log \det M)| \le C \tag{4}$$

uniformly in K, preceding depth, and the chosen joint structural systems. The coefficient is the actual affine Maxwell coefficient.

1.1 Exact algebra and uniform flat estimates

Choose an orthonormal frame V for $\ker B$ on real backgrounds. In the frame $[V, J]$, the upper block of T is $G_B = V^T \Delta V$. Schur complementation gives

$$\det G_B = \det T \, \det(J^T T^{-1} J).$$

Since $\det \widehat{G} = (\mu^2)^{\dim S_c} \det G_B$, this proves (3). Positivity at flatness fixes the logarithm branches; holomorphic uniqueness continues the identity. This is finite determinant algebra, with no separately singular globally based scalar factor.

Write a subscript 0 for flatness. Internal-block Neumann Poincare gives

$$\|P_0 \phi\|^2 \le C_P K^2 \|D_0 \phi\|^2 .$$

Only internal block edges are used, on which a block constant has zero derivative. Consequently

$$c(\Delta_0 + \mu^2 I) \le T_0 \le \Delta_0 + \mu^2 I, \qquad c = \big(1 + \max(C_P, 1)\big)^{-1}. \tag{5}$$

There is no derivative of a sharp block indicator in this argument.

The product-tent lift from Notebook 35, composed with J_0, gives a map $E : S_c \to S$ satisfying

$$J_0^T E = I, \qquad \|E\| \le C_0, \qquad \|D_0 E\| \le C_1 \mu.$$

Let S be the minimum energy with prescribed $J_0^T \phi = b$. The feasible field Eb proves $0 \le S \le C_1^2 \mu^2 I$. The lower inverse block then gives

$$M_0 = (I + \mu^{-2} S)^{-1}, \qquad (1 + C_1^2)^{-1} I \le M_0 \le I. \tag{6}$$

At $K = 1$, the same statement uses the identity lift and the bounded unit-lattice gradient.

1.2 Uniform physical domain and weighted inverses

The established ordinary-reference trajectory bounds give intermediate angles $O(\ell^j \|A\|_\infty / K)$. Their sum over $j < k$ is bounded by $C \|A\|_\infty$. The hierarchical adjoint transports defining J therefore satisfy, on a fixed complex physical ball,

$$\|J(A) - J_0\| \le C \|A\|_\infty . \tag{7}$$

This is an operator-valued block bound: normalizing the K^4 entries by K^{-2} removes the counting dimension. Block Banach Cauchy gives the same uniform first-two derivative bounds. Only the physical supremum norm is needed for this scalar statement.

The local gauge generator obeys $\|D(A) - D_0\| \le C\mu \|A\|_\infty$. Expanding its product and using (5) gives

$$\left\| T_0^{-1/2} (T(A) - T_0) T_0^{-1/2} \right\| \le C(\|A\|_\infty + \|A\|_\infty^2). \tag{8}$$

The penalty difference is $\mu^2(\Pi(A) - \Pi_0)$, with the same bound. This is a complex operator-norm estimate relative to positive real T_0, not an assertion that a complex transpose product is positive. A Neumann inverse and (7) imply

$$\left\| T(A)^{-1} \right\| \le C\mu^{-2}, \qquad \|M(A) - M_0\| \le C \|A\|_\infty .$$

Together with (6), these give a common radius independent of K.

Conjugate by $e^{\theta f(x/K)}$, where f is real and one-Lipschitz. Use initial-site weights on bonds and center weights on coarse sites. The covariant difference changes by $O(\theta\mu)$; a J column has bounded physical diameter. The conjugated transpose carries the opposite weight. The same relative Neumann argument, with $C(\theta + \|A\|_\infty)$, proves for $R = T^{-1}$

$$\|R\|_\theta \le C\mu^{-2}, \qquad \|DR\|_\theta + \left\| RD^T \right\|_\theta \le C\mu^{-1}, \qquad \left\| DRD^T \right\|_\theta \le C. \tag{9}$$

For example, place $T_0^{-1/2}$ on the two sides of the Neumann inverse and use $\left\| D_\theta T_0^{-1/2} \right\| \le C$. The bounds hold between restricted final blocks, with an extra $e^{-c\, d(x,y)}$. Thus

$$\|M_{xy}\| \le C e^{-c\, d(x,y)}. \tag{10}$$

The coarse fiber has fixed color dimension. Schur summation in (10) shows that a small coarse exponential conjugation changes M_0 by $O(\theta)$. Weighted $M(A) - M_0$ is $O(\|A\|_\infty)$. A common logarithm contour and its resolvents therefore give uniform exponential kernels for $\log M$.

1.3 Rooted derivatives and the actual affine coefficient

Treat physical directions in one final block as one supremum-norm Banach space. A local direction a_j satisfies

$$\|\delta_j D\| \le C\mu \|a_j\| , \qquad \|\delta_j J\| \le C \|a_j\| .$$

The second primitive derivatives have the same respective powers or better and joint local support. Differentiate

$$\delta R = -R(\delta T)R, \qquad \delta T = (\delta D)^T D + D^T \delta D + \mu^2((\delta J)J^T + J(\delta J)^T).$$

Every term in the first-two derivatives of $M = \mu^2 J^T R J$ has balanced powers of μ, by (9). For example,

$$\mu^2 (J^T R D^T)(\delta D)(RJ) \quad \text{costs} \quad \mu^2 \mu^{-1} \mu \mu^{-2} = 1.$$

At two insertions, the intervening pieces are R, DR, RD^T, or DRD^T, with exactly the powers in (9). Differentiating J gives local endpoint terms; a penalty insertion supplies μ^2. The number of terms is independent of K.

Insert block projections at the differentiated factors. The restricted estimates connect every source and matrix endpoint. For $n = 1, 2$, after reducing the exponent,

$$\|\delta_{j_1} \cdots \delta_{j_n} M_{xy}\| \le C_n \prod_{\ell=1}^{n} \|a_{j_\ell}\| \; e^{-c\,\tau(x,y,j_1,\dots,j_n)}. \tag{11}$$

Here τ is minimum tree length on the final-block graph. The differentiated contour formula for $\log M$ preserves this estimate: its intermediate sums are over fixed-dimensional coarse fibers, and an exponential reserve pays them. This proves rooted second-derivative bounds with every fixed polynomial source moment. It does not sum individual fine-coordinate derivative norms.

By covariance, $\rho_y = \mathcal{N}_0 \operatorname{tr}(\log M)_{yy}$ is individually gauge invariant, and its linear color jet vanishes. At each fixed K, the actual stationary polynomial-jet theorem identifies its affine source with

$$A_\alpha(x) = t(x_\nu/K)H, \qquad \alpha \neq \nu,$$

up to a fine gauge. Its fine link angle is $tx_\nu H/K^2$. In source block j the coefficient of t has norm at most $C(1 + |j|)$. Inserting these directions in the second version of (11) for $\log M$ gives a uniformly finite sum. The invariant-density Ward identity of Notebook 35 identifies $\beta = 2[t^2]\rho_0$, proving (4). This uses the fixed-K actual identification and then estimates the explicit representative; it assumes no uniform stationary response as K grows.

It follows that

$$\beta(\mathcal{N}_0 \log\det \widehat{G}) = \beta(\mathcal{N}_0 \log\det T) + O(1). \tag{12}$$

Equality of total finite determinants and localized Hessians is used here. Their chosen diagonal densities need not agree pointwise. The remaining magnetic comparison for T is separate: a dimension-blind Cauchy bound on its fine trace can still grow with K, and does not compute its logarithmic coefficient.

2 Covariance and physical relative Hessian vertices

This section controls the first two background coefficients of an actual action Hessian. It does not estimate the moving hard mean-zero projection, nor compute a logarithmic determinant coefficient. The argument uses exact gauge covariance before taking absolute values.

2.1 Hypotheses and conclusion

Work on the current unit lattice with counting inner products and a fixed finite bond/color fiber. Let $L(a)$ be a transpose-symmetric family of left-relative bond Hessians. Assume that it is exactly covariant under sitewise adjoint frames and that its current-coordinate Taylor kernels through background degree two at zero have uniform exponentially rooted absolute row bounds. The transposed bounds are included. For an action Hessian, these are the action kernels through degree four together with the local relative-coordinate changes.

Write $q = L(0)$. Assume that q is translation invariant, scalar on each color factor, and exponentially summable, uniformly in the old classical depth. In source-site coordinates impose the componentwise moment conditions

$$\sum_r q_{\alpha\beta}(r) = 0, \qquad \sum_r r_\rho q_{\alpha\beta}(r) = 0 \quad \text{for every } \alpha, \beta, \rho. \tag{13}$$

Thus its Fourier symbol vanishes to order two. No symmetry of $q_{\alpha\beta}(r)$ in α, β at a fixed r is assumed. Only the full transpose relation $q_{\alpha\beta}(r) = q_{\beta\alpha}(-r)^T$ is used.

The actual application here is on a chosen infinite structural system, where these continuous-momentum moments have been established. Explicit periodizations of that system are also covered with the specified lifted paths. An arbitrary actual finite-regulator winding kernel is not silently identified with such a periodization.

Set $\mu = K^{-1} \le 1$. For a complete physical field A, define

$$\|A\|_{\mathrm{phys},1} = \|A\|_\infty + \mu^{-1} \max_{\substack{e,e' \text{ nearest} \\ \text{same component}}} |A(e') - A(e)|. \tag{14}$$

Let $L_j = [t^j]L(t\mu A)$, $j = 1, 2$, and suppose $\|A\|_{\mathrm{phys},1} \le a_*$. Here D_0 denotes the ordinary componentwise gradient of a *bond field*, not the scalar gauge generator. Put $R_0 = D_0^T D_0 + \mu^2 I$.

Theorem 2.1. *Under these hypotheses there are operators Z_j, E_j such that*

$$L_j = D_0^T Z_j + Z_j^T D_0 + E_j, \qquad j = 1, 2, \tag{15}$$

with

$$\begin{aligned} \|Z_1\| &\le C\mu a_*, & \|E_1\| &\le C\mu^2 a_*, \\ \|Z_2\| &\le C\mu^2 a_*^2, & \|E_2\| &\le C\mu^2 a_*^2. \end{aligned} \tag{16}$$

Consequently

$$\|R_0^{-1/2} L_j R_0^{-1/2}\| \le C_j a_*^j, \qquad j = 1, 2. \tag{17}$$

The constants are independent of K and of the old depth. Mixed second coefficients follow by polarization. For directions supported in fixed smooth physical collars, the factors admit exponentially rooted final-block operator bounds with the same powers of μ.

For the actual homogeneous classical action S_M, the hypotheses come from the established uniform support and Cauchy bounds, gauge covariance, and the flat normalization

$$q_{M,\mathrm{mid}}(p) = |p|^2 I - pp^T + O(|p|^4).$$

The analytic midpoint-to-source-site phases preserve the zero and first jets, so they imply (13). This last input is stronger than row-sum cancellation alone.

2.2 Formal local path gauges

For each displacement r, choose a translation-covariant coordinate path of length $|r|_1$, with the negative-displacement path defined by reversal. Denote its adjoint transport by $W_{xy}(a)$. It obeys $W_{yx} = W_{xy}^T$ and is complex orthogonal coefficientwise.

Fix the source site x of a matrix row. The transports from x to every z define a formal site gauge $g_x(z)$, with $g_x(x) = I$. A transformed edge is a loop consisting of the two selected paths and that edge. Its length is at most $C(1 + |z - x|)$; a plaquette filling uses at most $C(1 + |z - x|)^2$ plaquettes. Write its link-angle jet as

$$b(t, z) = tb_1(z) + t^2 b_2(z) + O(t^3).$$

Discrete Stokes at first order and the physical first-difference bound give

$$|b_1(z)| \le C\mu^2 a_* (1 + |z - x|)^2. \tag{18}$$

The second coefficient of the product and its logarithm is a sum of quadratic pairs of first link coefficients. There are at most the square of the loop length such pairs. Hence

$$|b_2(z)| \le C\mu^2 a_*^2 (1 + |z - x|)^2. \tag{19}$$

These are formal coefficient identities. The potentially unbounded gauge g_x is not placed in a global analytic ball. One first truncates its polynomial jets far from x, applies the finite covariance identity, and then passes coefficientwise. The exponentially rooted bounds absorb the polynomial tails. In particular this step does not assert a finite-period identity for an unbounded affine field.

Gauge covariance and $g_x(x) = I$ give, with a consistent choice of transport orientation,

$$L(t\mu A)_{x\alpha,y\beta} = L(b(t))_{x\alpha,y\beta} W_{xy}(t\mu A). \tag{20}$$

The inverse convention gives the same argument with both orientations reversed. Define the dressed flat kernel

$$L^P(t)_{x\alpha,y\beta} = q_{\alpha\beta}(y-x) W_{xy}(t\mu A).$$

Using (18)–(19) and the rooted Taylor kernels,

$$\begin{aligned} \|[t](L - L^P)\|_{\mathrm{Schur}} &\le C\mu^2 a_*, \\ \|[t^2](L - L^P)\|_{\mathrm{Schur}} &\le C\mu^2 a_*^2. \end{aligned} \tag{21}$$

At second order the terms are $L'[b_2]$, $\frac{1}{2}L''[b_1, b_1]$, and the product of $L'[b_1]$ with the first transport coefficient. The latter two have, respectively, the stronger powers μ^4 and μ^3. All sums are fine current-coordinate Schur sums; no final-block fiber dimension is inserted.

2.3 Ordered factorization and covariant completion

A common difference-square formula would be invalid here: the product of two equal-displacement differences is symmetric in the bond components and would discard the odd-displacement, skew-component part of $q(r)$. Even the forward-gradient Maxwell kernel has such shifted entries. Instead use an ordered Laurent factorization.

For a Laurent series F, define

$$\mathcal{B}_j F(z) = \frac{F(1, \dots, 1, z_j, \dots, z_d) - F(1, \dots, 1, 1, z_{j+1}, \dots, z_d)}{z_j - 1}.$$

The quotient is removable. Applying the telescoping identity twice, and using (13), gives

$$q(z) = \sum_{\rho,\tau} (z_\rho^{-1} - 1) H_{\rho\tau}(z)(z_\tau - 1), \qquad H_{\rho\tau}(z) = -z_\rho \mathcal{B}_\tau \mathcal{B}_\rho q(z). \tag{22}$$

Each monomial division costs at most $|r|_1$ terms with no increase in exponent length. Spending an exponential reserve twice therefore yields a uniform exponentially summable H. For example an input exponent σ gives an output exponent $\sigma/2$ and a bound

$$\|H\|_{\sigma/2} \le \frac{16 e^{\sigma/2}}{e^2 \sigma^2} \|q\|_\sigma$$

in the combined absolute row norm; the analogous column bound follows in the same way. Replace H by $(H + H^T)/2$, where the transpose includes displacement reversal and both gradient and bond indices. The factorization is unchanged. We have

$$q = D_0^T H_0 D_0, \qquad H_0 = H_0^T. \tag{23}$$

The color-scalar property is preserved.

Dress each H_0 entry by its endpoint transport:

$$H^P(a)_{x\rho\alpha,y\tau\beta} = (H_0)_{\rho\alpha,\tau\beta}(y-x) W_{xy}(a).$$

Let D_a be the nearest-neighbor covariant component gradient and define

$$Q_{\mathrm{cov}}(a) = D_a^T H^P(a) D_a. \tag{24}$$

This family is exactly covariant and transpose symmetric, with $Q_{\mathrm{cov}}(0) = q$. Its first two Taylor kernels have uniform exponentially rooted bounds: path derivatives only add polynomial length factors. On a sufficiently small complex current-angle ball, the path-product factor $\exp(C\|a\|_\infty |r|)$ is paid by the exponential reserve.

Apply the local-gauge argument to Q_{cov} as well as to L. They have the same flat q, hence the same dressed flat comparison L^P. Subtraction of their two estimates (21) gives symmetric errors

$$\mathcal{E}_j = [t^j](L(t\mu A) - Q_{\mathrm{cov}}(t\mu A)), \qquad \|\mathcal{E}_j\|_{\mathrm{Schur}} \le C\mu^2 a_*^j, \quad j = 1, 2. \tag{25}$$

This argument does not presume that the paths generated by (24) coincide with the initially selected endpoint paths.

2.4 Coefficient expansion and localization

Write, through second order,

$$D_{t\mu A} = D_0 + tF_1 + t^2 F_2, \qquad H^P(t\mu A) = H_0 + tH_1 + t^2 H_2.$$

Nearest-neighbor differentiation and exponential path moments give

$$\begin{aligned} \|F_1\| + \|H_1\| &\le C\mu a_*, \\ \|F_2\| + \|H_2\| &\le C\mu^2 a_*^2. \end{aligned} \tag{26}$$

All H_j are symmetric. Set

$$\begin{aligned} Z_1 &= \tfrac{1}{2} H_1 D_0 + H_0 F_1, \\ Z_2 &= \tfrac{1}{2} H_2 D_0 + H_1 F_1 + H_0 F_2, \\ E_1 &= \mathcal{E}_1, \qquad E_2 = F_1^T H_0 F_1 + \mathcal{E}_2. \end{aligned} \tag{27}$$

Direct expansion of (24) proves (15). The counting norm of D_0 is bounded by a dimensional constant, so (26) and (25) prove (16). Finally,

$$\|D_0 R_0^{-1/2}\| \le 1, \qquad \|R_0^{-1/2}\| \le \mu^{-1},$$

which proves (17).

For localization, use complete directions supported in a fixed smooth physical partition with bounded overlap and bounded physical first derivatives. The terms F_1, F_2 have nearest-neighbor support. Each marked source in H_1, H_2 lies on a selected endpoint path. A distant source forces a long path, and the exponential H_0 kernel pays its length. In the remainders, a source on a local-gauge path is connected to the fixed matrix row through the original background-source endpoint. The original rooted Taylor bound pays that distance. Discrete Stokes uses source differences in the same smooth collar, enlarged by one fine edge. Polynomial path multiplicities are absorbed by a smaller exponential exponent.

Thus the factors in (27) and the E_j have rooted final-block bounds of the form $\exp[-c\tau(\text{row}, \text{column}, \text{source labels})]$ with the powers in (16). Fine-coordinate Schur summation proves the block-operator estimate; summing K^4 independent entry bounds is unnecessary. No sharp source cutoff is differentiated.

Physical weights $\exp(\theta f(x/K))$, with f Lipschitz, change a path factor by at most $\exp(C\theta\mu|r|)$, again paid by the exponential reserve. Commuting D_0 past the weight produces $O(\theta\mu)$. When multiplied by Z_1 or Z_2, this has the zero-order size required for E_1 or E_2. This supplies the weighted localized form of the theorem.

2.5 Scope of the actual application

For $L = S''_M$, the hypotheses are the uniform action support and Cauchy bounds through degree four, actual gauge covariance, and the already normalized flat kernel. The result concerns bounded physical C^1 fields and their localized jets. It does not give the same gain for rapidly oscillating bounded potentials, whose fine curvature may be $O(K^{-1})$, and it gives no estimate for the hard-projection scalar operator.

In the intended soft-vector application, the exact on-shell identification and its source-coordinate conventions must first justify the relevant affine jets of

$$S''_M(e^{iA/K}) + D_A D_A^T + \overline{C}_A^T[aI - S''_{MK}(M_K(e^{iA/K}))]\overline{C}_A.$$

The corrected constraint's full $O(K^{-1})$ bound and its physical source derivatives are separate inputs. The present theorem supplies the action-Hessian relative vertices. It does not by itself identify a soft-vector determinant, control every stationary response, or determine a one-loop sign.

3 A soft vector completion with a bounded coarse remainder

The exact identities below use the actual mean-zero determinant. The large-K estimate concerns its first two affine background jets. Its auxiliary off-shell family retains the effective action S_M; it is not replaced by a bare Wilson Hessian.

3.1 Exact finite factorization

Use current counting coordinates, $K = \ell^k$, $\mu = K^{-1}$, and

$$J = K^2 B^T, \qquad J^T J = I, \qquad \Pi = JJ^T, \qquad P = I - \Pi. \tag{28}$$

The actual covariant pair and residual Hessian are

$$\widehat{G} = PD^T DP + \mu^2 \Pi, \qquad Q_B = \mathcal{H} + DPD^T + a\bar{C}^T\bar{C}, \qquad \mathcal{H} = L - C^T qC. \tag{29}$$

Here $L = S''_M - \lambda \cdot f''$, $q = S''_{MK}$, all Hessians are left-relative, and $a \geq R_q + 1$ is fixed. The actual correction is $\mathcal{D} = \mathcal{N}_0(\log\det \widehat{G} - \frac{1}{2}\log\det Q_B)$. Define its soft vector completion and coarse Schur factor by

$$V = \mathcal{H} + DD^T + a\bar{C}^T\bar{C} = Q_B + (DJ)(DJ)^T, \qquad S = I - J^T D^T V^{-1} DJ. \tag{30}$$

The determinant lemma and Woodbury identity give

$$\det Q_B = \det V \det S, \qquad S = (I + J^T D^T Q_B^{-1} DJ)^{-1}. \tag{31}$$

These are identities between positive matrices on the real small germ, with normalized holomorphic continuations through flatness.

For the soft scalar matrices

$$T_s = D^T D + \mu^2 \Pi, \qquad M_s = \mu^2 J^T T_s^{-1} J, \tag{32}$$

the scalar Schur identity is $\det \widehat{G} = \det T_s \det M_s$. Thus the exact actual decomposition is

$$\boxed{\mathcal{D} = \mathcal{N}_0\left[\log\det T_s - \tfrac{1}{2}\log\det V + \log\det M_s - \tfrac{1}{2}\log\det S\right].} \tag{33}$$

No massless based scalar factor or omitted Haar normalization occurs. The matrix S has one fixed color fiber per final scalar site and is covariant in that coarse frame.

3.2 Uniform flat spectra and energy resolvents

A subscript 0 denotes flat background. The proved mean-zero estimates are $\Delta_1 \le C_\Delta Q_{B,0}$ and $Q_{B,0} \ge c\mu^2 I$. Since $D_0\Pi_0 D_0^T \le D_0 D_0^T \le \Delta_1$,

$$Q_{B,0} \le V_0 \le (1+C_\Delta)Q_{B,0}, \qquad (1+C_\Delta)^{-1} I \le S_0 \le I. \tag{34}$$

Indeed $\|Q_{B,0}^{-1/2} D_0 J_0\|^2 \le C_\Delta$ and (31) give the second assertion. The number of coarse rows has not been used as an estimate for this condition number.

At zero, $qD_c = 0$, so the ordinary and corrected constraint terms agree inside $C^T qC$. With $A_K = \bar{C}(0)$,

$$V_0 = q_M + D_0 D_0^T + A_K^T (aI - q_{MK}) A_K \ge \Delta_1 + A_K^T A_K. \tag{35}$$

The uniform analytic symbol of q_M vanishes to order two at zero. Its uniform second moments therefore give $q_M \le C\Delta_1$. Also $\|A_K\| \le \mu$ and $\|q_{MK}\| \le R_q$. Combining these facts with the gap gives

$$c(\Delta_1 + \mu^2 I) \le V_0 \le C(\Delta_1 + \mu^2 I), \tag{36}$$

on the chosen infinite structural system and its explicit periodizations. The finite actual lower bound, gap and heat estimate below remain valid without identifying arbitrary finite winding kernels with those periodizations.

For the spatial argument, q_M is the kernel of the chosen infinite structural system. The same conclusions hold for its explicitly periodized kernels; they are not asserted for unrelated finite winding kernels merely by sampling. For later source estimates the spatial exponent must also be uniform in final units. If $\delta(p)$ is the scalar lattice symbol, the common analytic strip and double zero of q_M imply

$$\|q_M(p + i\mu\theta) - q_M(p)\| \le C\{\mu|\theta|\sqrt{\delta(p)} + \mu^2|\theta|^2\}. \tag{37}$$

Near zero this is Taylor's formula; away from zero it follows from bounded derivatives. The ordinary longitudinal term has the same bound. Conjugating fine fields by $e^{\theta\cdot x/K}$ and coarse fields by $e^{\theta\cdot y}$, straight-average row and column counting gives

$$\|A_{K,\theta}\| \le C\mu, \qquad \|A_{K,\theta} - A_K\| \le C|\theta|\mu. \tag{38}$$

The coarse symbol q_{MK} has a uniform strip. Its coarse correction therefore changes by $O(|\theta|\mu^2)$. Measured between two $(\Delta_1 + \mu^2 I)^{-1/2}$ factors, the total change is $O(|\theta|)$. Equations (36)– (38) give a common physical strip on which all energy sandwiches of V_0^{-1} are bounded uniformly. The energy legs are the ordinary componentwise first differences and μI.

Bloch decomposition by K-blocks turns these into bounded operator-valued analytic functions. Fourier integration gives uniform exponentially summable final-block kernels. The internal fiber may have dimension K^4, but the bounds are Hilbert operator norms; no entrywise sum over that fiber is taken. Explicit periodizations obey the same bounds. The established fixed-K local-limit interface identifies the resulting infinite source coefficient with the actual local Maxwell coefficient; it does not identify arbitrary finite kernels with these periodizations. In particular $D_0^T V_0^{-1} D_0$, S_0 and, after shrinking the exponent using (34), S_0^{-1} have uniform weighted block bounds.

Min–max also gives the flat heat estimate

$$N^{-4} \operatorname{Tr} e^{-uK^2 V_0} \le C \left\{ \frac{K^4}{(1+K^2u)^2} + N^{-4} \right\} e^{-cu}. \tag{39}$$

It follows from $V_0 \ge Q_{B,0}$ and the mean-zero heat theorem. The same Mellin integration gives bounded inverse moments for powers $p > 2$, the logarithmic bound at $p = 2$, and exponential final-time tails. These flat estimates alone say nothing about a magnetic derivative.

3.3 The auxiliary family and the exact affine matching

For bounded physical C^1 potentials A put $U_A = e^{i\mu A}$ and

$$\widetilde{V}(A) = S''_M(U_A) + D_A D_A^T + \bar{C}_A^T\{aI - S''_{MK}(M_K(U_A))\}\bar{C}_A, \tag{40}$$

$$\widetilde{S}(A) = I - J_A^T D_A^T \widetilde{V}(A)^{-1} D_A J_A. \tag{41}$$

Every Hessian in these formulas is the actual relative-coordinate classical Hessian. The family is covariant, but it is not declared equal to (30) off shell.

On the established commuting affine probe, the actual fine and coarse currents and the multiplier vanish through the two jets used in the Maxwell extraction. Thus $qD_c = 0$ and

$$C^T q C = \bar{C}^T q \bar{C}, \qquad L = S''_M \quad \text{through the first two affine jets.} \tag{42}$$

The actual smooth stationary polynomial jets are the corresponding fine affine connection. Hence V, S and $\widetilde{V}, \widetilde{S}$ have identical first two affine jets. This is the precise interface used below.

3.4 Relative vertices with localized sources

Use complete physical source fields with

$$a_* = \|A\|_\infty + \mu^{-1} \max_{\text{nearest, same component}} |A(e') - A(e)|.$$

The corrected relative-Hessian theorem supplies, for $j = 1, 2$,

$$\begin{aligned} [t^j]S''_M(e^{it\mu A}) &= d_0^T Z_j + Z_j^T d_0 + E_j, \\ \|Z_1\| &\le C\mu a_*, & \|E_1\| &\le C\mu^2 a_*, \\ \|Z_2\| &\le C\mu^2 a_*^2, & \|E_2\| &\le C\mu^2 a_*^2. \end{aligned} \tag{43}$$

Here d_0 is the componentwise gradient of a bond field. These are source-rooted block estimates, including physical weights. Their proof uses the ordered Laurent factorization $q_M = d_0^T H_0 d_0$ and a symmetric covariant completion. A common componentwise-difference formula would be false for the shifted component-skew kernel and is not used.

The first two D_A jets are multiplication operators of sizes $O(\mu a_*)$ and $O(\mu^2 a_*^2)$. Their contribution to $D_A D_A^T$ has the same form as (43). The actual moving-reference products obey $\|\bar{C}_A\| = O(\mu)$ with first two physical-background derivatives of the same scale. Their rows and sources lie in two final blocks. The ordinary output $M_K(e^{i\mu A})$ is uniformly analytic and $O(\|A\|)$ in final units, with the same fixed physical range. The previously proved relative coarse Hessian kernels are uniform in MK. Therefore the last term of (40), including its first two derivatives, is a zero-order $O(\mu^2)$ vertex with exponentially rooted sources. These conclusions use corrected moving tangents, not an estimate for the full uncorrected ordinary adjoint.

The normalized block map J_A and its first two physical derivatives have uniform own-block operator bounds. Keep the factorization

$$R(A) = D_A^T \widetilde{V}(A)^{-1} D_A, \qquad \widetilde{S}(A) = I - J_A^T R(A) J_A. \tag{44}$$

The derivatives of J stay outside R. In particular, no raw derivative of DJ at a sharp block interface is estimated before using a resolvent. Since $V_0 \ge D_0 D_0^T$ and $V_0 \ge c\mu^2 I$,

$$\|V_0^{-1/2} D_0\| \le 1, \qquad \|V_0^{-1/2} D_0'\| \le C a_*. \tag{45}$$

For example,

$$R' = D'^T V_0^{-1} D_0 + D_0^T V_0^{-1} D' - D_0^T V_0^{-1} V' V_0^{-1} D_0. \tag{46}$$

Differentiating once more yields the ordered two-V' terms, one V'' term, and the D'' and mixed D', V' terms. Insert (43) between the uniform weighted energy resolvents. Each term is bounded with its physical source norms, by (45); this proves both first and second response bounds without a fine-coordinate dimension factor.

For unit directions supported in smooth physical collars x, y, a spare exponent attaches the sources to their resolvent legs and gives

$$\begin{aligned} \|\widetilde{S}'_x(i,j)\| &\le C e^{-m\tau(i,j,x)}, \\ \|\widetilde{S}''_{x,y}(i,j)\| &\le C e^{-m\tau(i,j,x,y)}. \end{aligned} \tag{47}$$

Here τ is connecting-tree length including every displayed operator endpoint and source. With a smaller exponent, these imply $\sum_{i,j} e^{m'[d(i,x)+d(j,x)]} \|\widetilde{S}'_x(i,j)\| \le C$ and $\sum_{i,j} \|\widetilde{S}''_{x,y}(i,j)\| \le C e^{-m' d(x,y)}$. The full rooted estimate, rather than only these summed consequences, is retained in the diagonal-density argument below. The indices are final-block labels, with fixed color fibers. The same source bounds hold after multiplying by J, J', J'', because these are normalized own-block operators. Completeness of the fine field collars is essential; sharp background cutoffs are not differentiated.

3.5 The coarse logarithm and its actual Maxwell bound

Equations (34) and (47) give for the total Hessian

$$K_S(x,y) = \operatorname{Tr}\left[S_0^{-1}\widetilde{S}''_{x,y} - S_0^{-1}\widetilde{S}'_x S_0^{-1}\widetilde{S}'_y\right], \qquad |K_S(x,y)| \le C e^{-m' d(x,y)}. \tag{48}$$

Both traces are coarse traces with fixed color dimension.

For the local diagonal density one must not cyclically rearrange a partial trace. Put $R_t = (S_0 + tI)^{-1}$. Its second source derivative at root zero is the partial trace of the integral over $t \ge 0$ of

$$R_t \widetilde{S}''_{x,y} R_t - R_t \widetilde{S}'_x R_t \widetilde{S}'_y R_t - R_t \widetilde{S}'_y R_t \widetilde{S}'_x R_t. \tag{49}$$

The weighted bound is $\|R_t\|_m \le C/(1+t)$. Thus its rooted density kernel is bounded by $C e^{-m''\tau(0,x,y)}$. This gives uniform polynomial source moments, not merely decay in the source separation.

Covariance makes each diagonal color trace invariant. Apply the rooted bounds to a smooth physical partition of an affine potential; the collar norms grow only linearly with their distance from zero and are summable against the rooted exponential. The density Ward and reanchoring identity then yields

$$\left|\beta\left(\tfrac{1}{2}\mathcal{N}_0 \log\det S\right)\right| \le C_S, \tag{50}$$

uniformly in M, K, in the actual joint infinite structural systems and polynomial-jet convention. The transfer from $\widetilde{S}$ to the actual S uses exactly (42). The linear color jet vanishes by semisimple conjugation invariance.

Combining (33) with the scalar coarse theorem gives

$$\beta(\mathcal{D}) = \beta\Big(\mathcal{N}_0[\log\det T_s - \tfrac{1}{2}\log\det V]\Big) + O(1), \tag{51}$$

with a remainder uniform in old depth and endpoint. For the vector coefficient, the auxiliary expression (40) is valid through its two affine jets. The fine soft pair has not yet been compared to a standard massive vector/ghost reference. A fixed-integrand universality theorem alone does not supply bounds uniform in this family. This module proves neither that remaining logarithmic comparison nor its sign, and it does not assert a full K-independent nonlinear background domain. Finite-volume winding and normalized-extraction issues retain their separate regulator hypotheses.

4 The fine short-time band of the soft pair

This argument is on the chosen infinite structural system, in current counting coordinates, with $\mu = K^{-1}$. All unbounded backgrounds are formal polynomial jets of the commuting affine probe

$$U_\alpha(t;x) = \exp(itx_\nu H/K^2), \qquad \alpha \neq \nu.$$

It asserts no continuously varying magnetic flux on a finite torus. The actual action S_M is retained.

Put

$$\begin{aligned} B_s(t) &= D(t)^T D(t), & B_v(t) &= S''_M(U(t)) + D(t)D(t)^T, \\ T_s(t) &= B_s(t) + \mu^2 J(t)J(t)^T, & \widetilde{V}(t) &= B_v(t) + \bar{C}(t)^T F(t)\bar{C}(t), \\ F(t) &= aI - S''_{MK}(M_K(U(t))). \end{aligned} \tag{52}$$

The scalar and vector final-block-zero projectors are denoted $P_{0,s}$ and $P_{0,v}$.

Theorem 4.1. *There is a constant independent of old depth M and endpoint K such that*

$$\begin{aligned} \int_0^1 \left| [t^2] \operatorname{Tr}(P_{0,s} e^{-sT_s(t)}) \right| \frac{ds}{s} &\leq C, \\ \int_0^1 \left| [t^2] \operatorname{Tr}(P_{0,v} e^{-s\widetilde{V}(t)}) \right| \frac{ds}{s} &\leq C. \end{aligned} \tag{53}$$

The vector expression equals the actual soft vector through its two affine jets. The effect of each soft coarse penalty alone is bounded by $C\mu^2 s$ before integration.

4.1 Rooted estimates for the base operators

At flatness B_s and B_v have uniform exponential fine-current Schur bounds. The vector statement uses the actual q_M; its first two background derivatives use the uniform action kernels through degree four and exact left-relative covariance. The gauge terms are nearest-neighbor. The semigroup power series therefore has Schur bound Ce^{Cs} for $0 \leq s \leq 1$.

At a diagonal fine root x, a Cartan gauge subtracts the constant potential at that root. The resulting source is $t(z_\nu - x_\nu)H/K^2$. Diagonal color trace is unchanged. Each first magnetic insertion now supplies K^{-2} times a polynomial in distances to the root, and each second insertion supplies K^{-4}. The rooted exponential kernels pay the polynomial factors. The ordered one-insertion and two-insertion heat formulas give, for $E = B_s, B_v$,

$$\left| [t^2] \operatorname{tr}_{\mathfrak{g}} (e^{-sE(t)})_{xx} \right| \leq CK^{-4}(s + s^2)e^{Cs}, \qquad 0 < s \leq 1. \tag{54}$$

There are K^4 scalar roots or dK^4 initial bond roots in one block. Thus

$$\left| [t^2] \operatorname{Tr}(P_{0,E} e^{-sE(t)}) \right| \leq C(s + s^2)e^{Cs}. \tag{55}$$

This proof uses no spectral gap. Polynomial gauges are justified by compact truncation and exponential absolute moments, not by evaluating an unbounded field inside a global analytic ball.

4.2 Fixed coarse fibers for the penalty insertion

Write either perturbation as $C(t)^T F(t) C(t)$, with

$$(C, F) = \begin{cases} (\mu J^T, I), & \text{scalar}, \\ (\bar{C}, aI - S''_{MK}(M_K(U))), & \text{vector}. \end{cases} \tag{56}$$

The output fiber of C has fixed color/site or color/bond dimension per final block. The actual moving-reference physical estimates, centered on each dependency collar, give for $j = 0, 1, 2$

$$\|C_j(Y, X)\| \le C\mu(1 + |X| + |Y|)^p \mathbf{1}_{d(X,Y)\le r_0}. \tag{57}$$

Here $C_j = [t^j]C(t)$, and r_0, p are fixed. For J the support is its own block. Centering the affine field uses bounded physical C^1 directions; restoring the gauge frames costs only a fixed polynomial in the block labels.

The ordinary map sends the affine probe to the coarse affine connection. The actual coarse Hessian source bounds consequently give

$$\|F_j(Y, Z)\| \le C(1 + |Y| + |Z|)^p e^{-\kappa d(Y,Z)}. \tag{58}$$

These are whole-block Hilbert operator bounds, with no sum over K^4 independent fine coordinates.

The base jets also have polynomially weighted exponential final-block operator kernels. Neighboring blocks are controlled by the fine Schur bound; separated blocks gain the fine exponential at distance of order K times the block distance. The flat semigroup series and its first two insertion formulas therefore give the same kind of bound, uniformly for $0 \le s \le 1$, for both B and $W = B + C^T FC$. This step takes no fine trace.

The exact Duhamel formula is

$$e^{-sW(t)} - e^{-sB(t)} = -\int_0^s e^{-(s-u)W(t)} C(t)^T F(t) C(t) e^{-uB(t)}\, du. \tag{59}$$

Keep the fine root projector in the trace:

$$\mathrm{Tr}_f[P_0 E_W C^T FCE_B] = \mathrm{Tr}_c[FCE_B P_0 E_W C^T]. \tag{60}$$

The right side sums all coarse diagonal blocks; it is not a single coarse block-zero density.

Expand its degree-two coefficient in the finitely many ordered ways. There are always two C factors, supplying μ^2. Equations (57)–(58) and the semigroup bounds connect both coarse endpoints to block zero. Their absolute sum is bounded by

$$C\mu^2 \sum_{Y,Z} (1 + |Y| + |Z|)^p e^{-\kappa'(|Y|+|Z|+d(Y,Z))} \le C'\mu^2.$$

The trace is taken in a fixed-dimensional coarse fiber. A product through the large fine block Hilbert space costs this coarse output rank, not the fine dimension. Integrating (59) gives

$$\left| [t^2] \,\mathrm{Tr}\, P_0 (e^{-sW(t)} - e^{-sB(t)}) \right| \le C\mu^2 s, \qquad 0 < s \le 1. \tag{61}$$

For compact source truncations, (60) is ordinary finite-rank trace cyclicity. The displayed rooted sum supplies its infinite limit and permits coefficient extraction. This also justifies the complex-bilinear continuation at the real flat expansion point.

Combining (55) and (61), dividing by s, and integrating proves (53). The exact affine matching of the soft vector transfers this estimate to the actual vector coefficient.

4.3 Scope

The band $0 < s < 1$ is $0 < u < K^{-2}$ in final physical heat time. Its contribution to the actual signed soft-pair coefficient is bounded uniformly. This proof does not compute the logarithmic coefficient on $1 < s < K^2$, compare the actual effective action to a standard massive vector/ghost reference, or establish a large-time cancellation. Finite winding and regulator identification retain their separate hypotheses.

5 A uniform affine infrared tail from local cubic resolvents

This section bounds the large-physical-time contribution of the actual soft scalar/vector pair. The proof retains the root block in every trace and introduces no factor counting its K^4 fine variables. It uses the localized relative vertices proved above, not only a global relative-form estimate. It does not determine the intermediate-time logarithmic coefficient.

5.1 Precise operator and source inputs

Write $\mu = K^{-1}$. Fine scalar or positive-bond spaces use counting inner products, with fixed color/component multiplicities. Assign a bond to the block containing its initial site. Let P_x be the orthogonal projection onto the final block x, and put

$$\mathcal{L} = I + K^2\Delta, \tag{62}$$

where Δ is the ordinary componentwise flat lattice Laplacian. Let X_0 denote the flat physical operator, namely K^2 times the corresponding counting matrix.

For the vector spatial estimates, work on the chosen joint infinite structural system or on explicit periodizations of its kernels. Actual finite-regulator gaps and flat heat estimates are separate facts. A finite momentum grid does not imply exact infinite moment cancellation, and arbitrary finite winding kernels are not identified with these periodizations. The following abstract proof is also finite-volume uniform whenever its three inputs hold on the finite system in question.

The first input is uniform positive ellipticity:

$$X_0 = X_0^*, \qquad X_0 \geq c_0\mathcal{L}. \tag{63}$$

The second is a relative physical-weight estimate. For every bounded physical 1-Lipschitz function f, set $W = e^{\theta f}$, using the fine-site position divided by K. On bonds use their initial sites. For sufficiently small fixed $\theta > 0$, assume

$$\left\| \mathcal{L}^{-1/2}(WX_0W^{-1} - X_0)\mathcal{L}^{-1/2} \right\| \leq C\theta. \tag{64}$$

Bounded truncations of distance functions suffice.

For the third input, let a_j be a complete physical C^1 field supported in a fixed smooth collar of block j, and write

$$a_j^* = \|a_j\|_{\mathrm{phys},1} = \|a_j\|_\infty + K \max_{\substack{\text{nearest} \\ \text{same component}}} |a_j(e') - a_j(e)|. \tag{65}$$

Denote the first and mixed second variations of X at zero by V_j and V_{jk}. For every weight allowed above, require

$$\begin{aligned} \|\mathcal{L}^{-1/2}WV_jW^{-1}\mathcal{L}^{-1/2}\| &\leq Ca_j^*, \\ \|\mathcal{L}^{-1/2}WV_{jk}W^{-1}\mathcal{L}^{-1/2}\| &\leq Ca_j^*a_k^*. \end{aligned} \tag{66}$$

In addition, let f_j be physical distance to the fixed physical cube or collar of block j, and $W_j = e^{\theta f_j}$, initially with bounded truncations. The marked-source bounds are

$$\begin{aligned} \|\mathcal{L}^{-1/2}W_jV_jW_j\mathcal{L}^{-1/2}\| &\leq Ca_j^*, \\ \|\mathcal{L}^{-1/2}W_jV_{jk}W_j\mathcal{L}^{-1/2}\| &\leq Ca_j^*a_k^*, \end{aligned} \tag{67}$$

with the second bound also centered at k. The function f_j is Lipschitz on fine physical positions: it is not a piecewise constant block-index distance. Its infimum and supremum on any one block differ by a dimensional constant.

The two-sided weights in (67) attach a source to both operator ends. An unlocalized bound on $X_0^{-1/2} V_j X_0^{-1/2}$ would not supply this input. All source jets may be holomorphic and bilinear. Only the flat operators $X_0, \mathcal{L}$ are required to be positive.

5.2 Why the actual soft pair supplies these inputs

For $T_s = D_A^T D_A + \mu^2 \Pi_A$, the soft scalar theorem gives (63). Its weighted D, J expansion gives (64). Its first two source derivatives give (66)–(67). Here J is normalized and own-block local, so its derivatives have no fine-fiber dimension loss. Scalar sources even need only their supremum norm.

For the soft vector, the mean-zero Laplacian domination and gap, preserved by the positive DJ update, imply (63) for $X_0 = K^2 V_0$. The relative-Hessian theorem gives, after physical rescaling, first-two source jets of the form

$$D_{\mathrm{phys}}^T Z + Z^T D_{\mathrm{phys}} + E, \tag{68}$$

with uniformly bounded, exponentially source-rooted factors Z, E. The second-order Z may have an additional beneficial factor μ. Since $\|D_{\mathrm{phys}} \mathcal{L}^{-1/2}\| \le 1$, these factors give the relative vertex bounds. For example,

$$W_j D_{\mathrm{phys}}^T Z W_j = D_{\mathrm{phys}}^T (W_j Z W_j) + [W_j, D_{\mathrm{phys}}^T] W_j^{-1} (W_j Z W_j).$$

The commutator in the second term is $O(\theta)$; source-rooted decay bounds $W_j Z W_j$ at a smaller exponent. This proves the two-sided normalization without commuting a fractional power of $\mathcal{L}$ through a weight.

The covariant $D_A D_A^T$ term has the same elementary structure. Each corrected angle average $\bar{C}_A$, and each of its first two physical-background derivatives, has counting size $O(\mu)$. Their pair therefore gives $O(\mu^2)$, canceled by K^2. The coarse Hessian has fixed color fiber and uniform source bounds. These terms supply zero-order vertices in (68). The soft-vector module separately identifies this auxiliary family with the actual stationary affine jets through order two, retaining the current and residual terms.

For completeness, the action contribution to (64) uses precisely the repaired ordered Laurent factorization

$$q_M = d_0^T H_0 d_0. \tag{69}$$

Both the constant and linear flat momentum jets vanish, and H_0 is symmetric with uniformly exponentially summable current kernels. Separate component row sums alone would not justify this step; no common componentwise difference-square identity is used. Conjugation changes each outer gradient by a bounded multiplication operator $O(\theta\mu)$, and changes H_0 by $O(\theta\mu)$ using its exponential first moment. Keep the latter change between the two gradients. After multiplication by K^2 and the two $\mathcal{L}^{-1/2}$ factors, the outer errors are $O(\theta + \theta^2)$ and the middle error is $O(\theta\mu)$. The longitudinal term obeys the same estimate. The bounded physical block correction has fixed physical range, with a coarse Hessian having uniform exponential moments, so its error is $O(\theta)$. This proves (64) without a false uniform bound on the absolute row norm of X_0, which can grow as K^2.

The uniform claims concern the explicit auxiliary source estimates. Their identification with the actual affine coefficient is used at each fixed K; no uniform stationary-response bound is needed.

5.3 A block Schatten bound without a rank loss

The elementary massive lattice heat estimate gives

$$\mathrm{Tr}(P_x \mathcal{L}^{-3} P_x) \le C \tag{70}$$

uniformly in K and compatible volume. Indeed, integrate the flat heat trace against $t^2e^{-t}/2$. On a torus of current side KN, the heat trace per final block is at most

$$C\left\{\frac{K^4}{(1+K^2t)^2}+N^{-4}\right\}.$$

Its indicated integral is bounded, including the finite zero mode. The infinite lattice obeys the same bound without that mode.

Compression is not interchanged with taking powers. If u_i is an orthonormal eigenbasis of $P_x\mathcal{L}^{-1}P_x$ in the block, scalar Jensen applied to the spectral measure of $\mathcal{L}^{-1}$ gives $\langle u_i,\mathcal{L}^{-1}u_i\rangle^3\le\langle u_i,\mathcal{L}^{-3}u_i\rangle$. Therefore

$$\|P_x\mathcal{L}^{-1/2}\|_{\mathfrak{S}_6}^6=\mathrm{Tr}[(P_x\mathcal{L}^{-1}P_x)^3]\le\mathrm{Tr}(P_x\mathcal{L}^{-3}P_x)\le C. \tag{71}$$

The analogous right-ended norm has the same bound. Schatten Hölder with exponents $6,\infty,6$ consequently yields

$$\|P_x\mathcal{L}^{-1/2}A\mathcal{L}^{-1/2}P_y\|_{\mathfrak{S}_3}\le C\|A\|. \tag{72}$$

There is no factor proportional to the block rank K^4.

5.4 A common contour and rooted resolvent derivatives

Choose $0<\lambda_0<c_0/4$ and $0<\varphi<\pi/2$. Let Γ be the positively oriented Dunford contour on the rays $\lambda_0+re^{\pm i\varphi}$, enclosing the positive spectrum to their right. At finite regulator, first close it beyond the largest eigenvalue. For $R_z=(z-X_0)^{-1}$,

$$\|\mathcal{L}^{1/2}R_z\mathcal{L}^{1/2}\|\le C_\Gamma,\qquad z\in\Gamma. \tag{73}$$

Indeed, $X_0\ge c_0\mathcal{L}$, while the spectral norm of $X_0^{1/2}R_zX_0^{1/2}$ is bounded on those rays. By (64) and a relative resolvent Neumann expansion, choosing θ sufficiently small gives

$$WR_zW^{-1}=\mathcal{L}^{-1/2}A_{z,W}\mathcal{L}^{-1/2},\qquad \|A_{z,W}\|\le C, \tag{74}$$

uniformly on the whole contour.

The first and mixed second derivatives are

$$\begin{aligned}\delta_jR_z&=R_zV_jR_z,\\ \delta_j\delta_kR_z&=R_zV_{jk}R_z+R_zV_jR_zV_kR_z+R_zV_kR_zV_jR_z.\end{aligned} \tag{75}$$

Relative normalization cancels adjacent powers of $\mathcal{L}$: each product is $\mathcal{L}^{-1/2}A\mathcal{L}^{-1/2}$ with uniformly bounded A.

Source localization can be checked without any fractional-weight commutator. For a marked source j,

$$W_j(R_zV_jR_z)W_j=(W_jR_zW_j^{-1})(W_jV_jW_j)(W_j^{-1}R_zW_j). \tag{76}$$

For the ordered term with another source k, put $R_+=W_jR_zW_j^{-1}$, $R_-=W_j^{-1}R_zW_j$; then

$$W_j(R_zV_jR_zV_kR_z)W_j=R_+(W_jV_jW_j)R_-(W_j^{-1}V_kW_j)R_-. \tag{77}$$

The marked vertex uses (67) and the other vertex uses (66). The primitive V_{jk} has the marked bound at either source. Thus (72) applies to every term.

Write τ for the shortest connected graph length in the final-block graph, allowing auxiliary vertices. The resulting block bounds, for $n = 0, 1, 2$, are

$$\|P_x \delta_{j_1} \cdots \delta_{j_n} R_z P_y\|_{\mathfrak{S}_3} \le C_n \prod_{\ell=1}^{n} a^*_{j_\ell} e^{-c\tau(x,y,j_1,\ldots,j_n)}. \tag{78}$$

For $n = 0$, use a one-sided distance weight from y. For $n = 1$, the marked weight gives $e^{-\theta[d(x,j)+d(y,j)]}$. For $n = 2$, take the geometric mean of the separate j- and k-centered estimates. Their four edges connect all marked vertices and dominate a connecting graph length. The infimum/supremum comparison of physical distances within each block changes only the constant. Fixed collars have the same effect. Periodic physical distances and bounded truncations give the identical conclusion whenever the inputs hold; the uniform bounds permit removal of the truncations.

5.5 Three resolvents preserve the local trace

Functional calculus gives, for $u > 0$,

$$e^{-uX} = \frac{2}{u^2} \frac{1}{2\pi i} \int_\Gamma e^{-uz} (z - X)^{-3} \, dz. \tag{79}$$

The scalar residue of the third-order pole is $u^2 e^{-u\lambda}/2$, fixing both coefficient and sign.

Differentiate R_z^3 through order two. Each term is a product of three differentiated resolvents, with derivative orders $n_1 + n_2 + n_3 \le 2$. Inserting the block resolution of the identity between them gives

$$\begin{aligned} &\left| \mathrm{Tr}(P_0 R_z^{(n_1)} P_i R_z^{(n_2)} P_j R_z^{(n_3)} P_0) \right| \\ &\qquad \le \|P_0 R_z^{(n_1)} P_i\|_{\mathfrak{S}_3} \|P_i R_z^{(n_2)} P_j\|_{\mathfrak{S}_3} \|P_j R_z^{(n_3)} P_0\|_{\mathfrak{S}_3}. \end{aligned} \tag{80}$$

Here superscripts denote the indicated source derivatives, not ordinary powers. The root block stays in place. Nothing is moved cyclically through P_0.

Equation (78) and exponential convolution, with a smaller exponent, imply

$$\left| \delta_{j_1} \cdots \delta_{j_n} \, \mathrm{Tr}(P_0 R_z^3 P_0) \right| \le C_n \prod_{\ell=1}^{n} a^*_{j_\ell} e^{-c'\tau(0,j_1,\ldots,j_n)}, \qquad n \le 2. \tag{81}$$

The two intermediate block sums are absolutely convergent in trace norm, uniformly in K, z and the applicable volume. This constructs the infinite local trace without an infinite extensive trace.

On the contour, $\mathrm{Re}\, z = \lambda_0 + r \cos\varphi$, whence

$$\int_\Gamma |e^{-uz}| \, |dz| \le C u^{-1} e^{-\lambda_0 u}.$$

Combining this with (79)–(81) proves

$$\left| \delta_{j_1} \cdots \delta_{j_n} \, \mathrm{Tr}(P_0 e^{-uX} P_0) \right| \le C_n u^{-3} e^{-\lambda_0 u} \prod_{\ell=1}^{n} a^*_{j_\ell} e^{-c'\tau(0,j_1,\ldots,j_n)}, \quad u \ge 1, \ n \le 2. \tag{82}$$

The displayed absolute majorants justify the differentiations, contour integral, block sums and local limits. At fixed K, begin with bounded compact sources in the actual local germ, then extend their coefficients using these bounds. No small-time asymptotic is claimed by the harmless factor u^{-3}.

5.6 The actual affine infrared contribution

Choose a smooth physical partition of unity on fixed block collars, with uniformly bounded first derivatives and finite overlap. Decompose the affine physical field

$$A_\alpha(x) = \frac{x_\beta}{K} H, \qquad \alpha \neq \beta, \tag{83}$$

into complete supported fields a_j. Their norms satisfy $a_j^* \leq C(1+|j|)$. The second source bound (82) is summable with these polynomial weights. Consequently

$$\left| [t^2] \operatorname{Tr}_0 e^{-uX(tA)} \right| \leq C e^{-cu}, \qquad u \geq 1, \tag{84}$$

uniformly in K and old depth on the stated structural systems. This is evaluation of exponentially localized Taylor kernels, not insertion of a nonzero unbounded field into a fixed analytic ball.

The actual first two stationary affine jets agree with the explicit representatives just estimated, up to a fine gauge. Their diagonal color traces are individually gauge invariant. Thus the fixed-K polynomial-jet identification transfers (84) to the actual coefficient before taking the uniform bound. It does not require a uniform-K norm for the full stationary response.

For $X_s = K^2 T_s$ and $X_v = K^2 V$, we obtain

Theorem 5.1 (Uniform actual infrared tail)**.** *Under the established soft-factor and localized relative-vertex inputs, the actual affine coefficient satisfies*

$$\int_1^\infty \left| [t^2] \left\{ \operatorname{Tr}_{0,E} e^{-uX_v(t)} - 2 \operatorname{Tr}_{0,S} e^{-uX_s(t)} \right\} \right| \frac{du}{u} \leq C. \tag{85}$$

The constant is uniform in K, old depth and the chosen joint infinite structural systems. Each factor separately obeys the same type of bound.

The fine link angle corresponding to (83) is $t x_\beta H / K^2$. The change of variables $s = K^2 u$ identifies (85) with the current-time tail $s \geq K^2$ of the actual soft-pair Maxwell heat formula. Constant determinant normalizations have zero affine coefficient.

No finite torus affine-flux family is asserted. Finite source bounds apply to explicit periodizations or to finite models for which the three inputs were separately verified. The actual polynomial coefficient is obtained in the chosen infinite structural system by the fixed-K identification; unrelated actual finite winding kernels are not silently substituted.

Together with the exact soft factorization and the bounded coarse Maxwell remainders, (85) removes the large-physical-time part from the logarithmic calculation. The remaining comparison is the short-current-time contribution and the intermediate range $1 \ll s \ll K^2$. Neither the value nor the sign of the logarithmic coefficient follows from this tail estimate.

6 The next comparison

The exact soft factorization and bounded coarse corrections reduce the required logarithm to the fine soft scalar/vector pair. Its short-current-time interval $0 < s < 1$ and its physical infrared tail $s > K^2$ both have uniformly bounded affine contributions. Writing

$$h^{\mathrm{soft}}_{M,K}(s) = [t^2] \left\{ \operatorname{Tr}_{0,E} e^{-sV(t)} - 2 \operatorname{Tr}_{0,S} e^{-sT_s(t)} \right\},$$

we therefore have the actual reduction

$$\boxed{\beta(\mathcal{D}_{M,K}) = \int_1^{K^2} h^{\mathrm{soft}}_{M,K}(s) \, \frac{ds}{s} + O(1),} \tag{86}$$

uniformly in preceding depth and endpoint on the stated structural systems. For $K = 1$ the intermediate interval is empty. The growing hard-projection vertices have been removed from this calculation by exact algebra. The actual effective-action Hessian remains present, with its now controlled relative physical vertices.

Two distinct ingredients remain in identifying the logarithm. First, the soft infrared penalties must be compared to a specified massive reference without discarding their gauge cancellations. Second, the actual effective-action vertices must be compared to the reference vertices with constants uniform in preceding depth. The previously proved nonlinear Ward extraction matches their marginal momentum jets through field degree four; a summable remainder estimate for the determinant is still required. A sufficient concrete target, with $\delta > 0$, is

$$\left| h^{\mathrm{soft}}_{M,K}(s) - 11\alpha_g \right| \le C \left(s^{-\delta} + (s/K^2)^{\delta} \right), \qquad 1 \le s \le K^2.$$

Its logarithmic-time integral gives the required $22\alpha_g \log K + O(1)$. This displayed comparison is a target, not an estimate proved in this notebook.

For a fixed logarithmically divergent one-loop lattice integral, Adams–Lee prove a universal logarithmic term with a finite remainder. Their theorem provides a useful reference-integral endpoint once its hypotheses and normalization are matched. It does not by itself apply uniformly to our growing block fibers and varying effective actions. The model-specific comparison must establish that interface.

A positive actual block drift would activate the finite transfer and tuning results in Notebook 32. The full existence argument also requires the recorded measure matching, mixed and large-field histories, sourced continuum limits, nontriviality, Euclidean covariance, positivity and reconstruction. These remain part of the original goal. The spectral gaps here are Gaussian fluctuation estimates, not a Yang–Mills mass-gap theorem.

References

[1] *Toward a four-dimensional Yang–Mills construction*, Notebooks 32–35 and the earlier classical support and Ward extraction modules in this workspace.

[2] D. H. Adams and W. Lee, *Structure of logarithmically divergent one-loop lattice Feynman integrals*, Phys. Rev. D **77** (2008), 045010, arXiv:0709.0781.

[3] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited accounts pinned at commit `e392ef45013a581664bab76b90867170b30fe10e`.